Exercise Sheet 2

Probability, Data and Estimation

Probabilistic Machine Learning

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2.A Conditioning and Bayes

Exercise 2.A.1: Consequences of the Axioms

Using only the three Kolmogorov axioms (and set algebra), show:

1. $P(A \setminus B) = P(A) - P(A \cap B)$
2. $P(A \cup B) \leq P(A) + P(B)$
3. If $A \subseteq B$, then $P(A) \leq P(B)$.

Exercise 2.A.2: One Flag, Two Flags

Fraud occurs in **0.2%** of transactions. A rule flags **95%** of frauds and **2%** of legitimate transactions.

1. What fraction of flagged transactions is fraud?
2. A second, different rule has the same rates. Assume the two rules err **independently given the class**. A transaction is flagged by both. Now?
3. When would the assumption in 2 be badly wrong — and in which direction would the answer then be off?

Exercise 2.A.3: Independent Is Not Disjoint

1. Show that two disjoint events A, B with $P(A), P(B) > 0$ are never independent.
2. Flip two fair coins. Let A = "first is heads", B = "second is heads", C = "the coins agree". Show that A, B, C are pairwise independent but not mutually independent.

2.B Random Variables and the Joint

Exercise 2.B.1: App Use and Churn

The joint distribution of X (uses the app: 1/0) and Y (churns: 1/0):

	$Y = 0$	$Y = 1$
$X = 0$	0.30	0.20
$X = 1$	0.40	0.10

1. Compute $P(X = 1)$, $P(Y = 1)$, $P(Y = 1 \mid X = 1)$ and $P(Y = 1 \mid X = 0)$.
2. Are X and Y independent? Compute $\mathrm{cov}(X, Y)$.
3. Does the table show that the app *reduces* churn? What else could produce it?

Exercise 2.B.2: Dependent, Yet Uncorrelated

Let X take the values $-1, 0, 1$ with probability $\frac{1}{3}$ each, and $Y = X^2$.

1. Compute $E[X]$, $E[Y]$ and $\mathrm{cov}(X, Y)$.
2. Show that X and Y are not independent.
3. What does this say about using correlation to screen features?

2.C Data and Statistics

Exercise 2.C.1: Scale and Model

For each attribute of a customer table, give the scale and a first probabilistic model, and name one statistic that would be **invalid**: payment method (card / invoice / PayPal) · delivery rating (1–5 stars) · number of orders last year · order value in € · has a newsletter subscription.

Exercise 2.C.2: How Precise Is a Conversion Rate?

A campaign reached **400** users; **60** converted.

1. Estimate the conversion rate and its standard error.
2. Give an approximate 95% confidence interval and state correctly what the 95% refers to.
3. How many users would give a half-width of ± 0.01?

Exercise 2.C.3: A Sampling Distribution by Simulation (Python)

```python
import numpy as np
rng = np.random.default_rng(0)
samples = rng.exponential(scale=2.0, size=(1000, 25))   # 1000 samples, n = 25
```

1. Compute the mean of each sample and the standard deviation of those means. Compare with $\sigma/\sqrt{n}$ (for this distribution $\sigma = 2$).
2. The exponential is strongly skewed. Plot a histogram of the means — what shape do you see, and which theorem predicts it?
3. Repeat with $n = 100$. By what factor does the spread shrink?